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AI-Driven Attendance Tracking with Haar Cascade and ResNet

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Abstract

An automated attendance system tracks attendance electronically, typically using facial recognition technology, streamlining record-keeping processes. This work presents a novel approach to attendance management utilizing advanced technologies such as the ResNet-50 algorithm and Haar Cascade for facial detection. The system leverages ResNet's deep learning architecture to extract unique facial features, combined with Haar Cascade's robust facial detection capabilities, resulting in efficient and accurate attendance recording and storing the attendance of all the previous days in an SQLite database. From the experimental results, it has been noted that even under occlusion situations, this model can identify nearly every face with 91.6% accuracy in low lighting and occlusion conditions.

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1. Introduction

In the rapidly changing digital age, moving from manual to automated attendance systems is no longer only a convenience; it is now a crucial step in enhancing operational accuracy, efficiency, and security across a wide range of businesses. Traditional attendance methods, such as punch cards and manual roll calls, are not only laborious but also prone to human error, which can result in vulnerabilities like proxy attendance and erroneous data. These old structures, which have been employed for a long time in business and educational contexts, often fail to meet the demands of modern organizational needs. They are inefficient and require a great deal of administrative effort, particularly in large-scale enterprises where accurate and current records are crucial. In order to streamline and modernize attendance management, institutions are increasingly implementing automated attendance systems that make use of cutting-edge technologies like facial recognition.[1]

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Automated attendance systems, for instance, have completely changed how student attendance is tracked in educational institutions. By ensuring accurate, automatic tracking of every student's attendance, these solutions reduce the likelihood of proxy attendance and ease the administrative load on teachers. Additionally, they offer administrators real-time insights regarding student attendance patterns, enabling them to keep an eye on attendance trends, deal with absenteeism, and make well-informed decisions. This greatly lessens the need for physical intervention while also ensuring adherence to attendance regulations and enhancing the general academic environment.[2]

In corporate settings, automated attendance systems are a vital tool for improving worker management. These systems ensure accurate and manageable data by automatically recording employee attendance. Through seamless integration with existing human resource (HR) management systems, they streamline payroll processes. [3]

The necessity to overcome the major drawbacks of traditional attendance tracking methods is the main motivation for the creation of an image processing-based real-time attendance marking system. In addition to being time-consuming, manual systems—which frequently depend on roll calls or sign-ins—are also very prone to human mistake and inaccuracies, which makes them ineffective for real-time tracking, particularly in large institutions. These approaches fall short of offering the instantaneous discernment required in contemporary settings where quick, trustworthy data is crucial. These issues are resolved by the suggested solution by utilizing cutting-edge facial recognition and picture processing technology. The system uses cameras or webcams to take pictures in the best possible lighting. The photos are then put through a number of preprocessing stages to improve their clarity and detail.[4] Then, to precisely identify facial regions from the image, advanced face identification algorithms like Convolutional Neural Networks (CNNs) or Haar cascades are used. Key face landmarks and traits are identified by analyzing these regions, and they are subsequently compared to a database of previously saved profiles. By extracting and assembling distinctive visual traits, the system can reliably identify an individual, automating the attendance process. The identified person's attendance is automatically tracked and saved, eliminating the need for human involvement and creating a seamless, up-to-date record. Because it not only increases accuracy but also ensures that attendance data is readily available for inspection and analysis, this is a scalable and efficient choice for institutions and businesses.

This work aims to improve and revolutionize the way attendance is managed by leveraging advanced technologies such as the resnet-50 algorithm. The photos of the attendees who are currently in class are first gathered, and the facial landmarks found using the Haar Cascade algorithm are then used as input to the ResNet algorithm to train the model. The recorded attendance is then stored in a SQLite database. The major contributions of this work are:

- A new Resnet model for this application has been built from scratch.
- A model to mark the attendance in occlusion conditions.
- This model also supports video processing. If the application isn't functioning at the moment, we can record a video and use it to mark attendance later.
- Accuracy, precision, recall, and the confusion matrix are utilized in performance analysis for evaluating performance.
- The results of this study are contrasted with the most recent SOTA models.

The paper is arranged in such a way that Section 2 brings together the recent literature on the attendance system, and Section 3 explores the proposed model used for the application. Section 4 throws light on the evaluation metrics used in this experiment, and Section 5 tells about the results obtained in the experiment and finally section 6 concludes remarks and offers insights for future research.

2. Literature Survey

This section explains the key research that has already been performed on the facial attendance system and is outlined as follows: S. Matilda et al. created a face detection and recognition model based on the Viola-Jones Haar cascade technique. The model maps each student's unique facial traits and obtains their details using a trained dataset. The model has an accuracy of 99.38% and provides clear face recognition results which can be stored in an Microsoft Excel.[5]

Ashesh Kumar et al. suggest an automatic attendance system that makes use of dlib for facial identification and OpenCV for frame extraction. With a faster reaction time, it can identify more faces from a single frame with greater

accuracy. The system facilitates students and teachers with a dashboard so that they can view their respective attendance reports.[6]

Karthikeyan et al. presented a system that uses the Local Binary Pattern Histogram (LBPH) algorithm with the Viola-Jones Face identification Technique for face identification and recognition. The video's frames are captured by the system, which then turns them into numbers. The human faces for the attendance record are then identified by comparing these figures with the preexisting data.[7]

A face recognition-based attendance system was employed by Komal Saxena et al. with the goal of replacing manual procedures and enhancing the precision of attendance tracking. The system effectively recognises pupils using facial identification algorithms such as Viola and Jones. [8]

Anupama Menon et. al. developed an AI-based student surveillance system that includes features like visualization modules for attendance data and email alert to the student and their ward when attendance falls below a minimum required level. The system incorporates machine learning algorithms for face recognition and attendance analysis.[9]

Setia Budi et. al. created an inexpensive technique to supervise student attendance. Simply identifying their faces on the attendance record—which is automatically created using face detection technology—allows students to register their attendance. Attendance is documented on class pictures.[10]

Mubarak Salem Mubarak Alburaiki et. al. suggested a framework that covered the following three key areas: First, use the cameras on mobile phones for automatic face detection and analysis. Second, a machine learning method is used by the facial recognition API. Maps API comes in third. More than thirty pupils have participated in a testing process aimed at determining how well the system recognises the faces and places of students.[11]

Azhaguraj.R. et.al. proposed an automated attendance control machine. The notion of transitioning from a traditional attendance method to virtual face recognition and identification techniques through the application of device learning algorithms has been deliberated.[12]

The authors highlight the necessity of creating unique models tailored to certain applications, which served as inspiration for their innovative CNN architecture for facial recognition. The proposed CNN architecture offers 99% accuracy, as shown by the experimental study reported in the paper.[13]

K. Vignesh and colleagues concentrated on creating a Convolutional Neural Network (CNN)-based facial recognition attendance management system. Their system also incorporates Open CV face recognition for image processing and attendance recording in an Excel spreadsheet.[14]

Kittipong Tapyou and the group used Raspberry Pi and Internet of Things devices to implement the HAAR Cascade and EBGM techniques. The quantity of faces or users in the system has no effect on the accuracy or distance. The range of average detecting distances was 306.68 cm to 416.67 cm, contingent upon the quantity of face photos utilised during the training phase.[15]

The main study reveals that the majority of algorithms like HAAR cascade, and Open CV were used to recognize faces: CNN and dlib to match the features of faces with accuracies such as 99%, 99.38%, etc. The attendance was marked into CSV files, Excel sheets, and dashboards.

Some gaps were observed in the key research such as the deep learning models that give results that are more accurate than the models used in research. The attendance data was stored in CSV files and Excel files, but an SQL database can be used which occupies less space. The previous works available are not able to recognise the face in low lighting conditions, different facial expressions, and occlusions (e.g., students wearing specs). In this research, we are using advanced deep learning models, and SQLite database to perk-up the model's performance compared to the previous work.

3. Proposed Model

A systematic approach has been developed for the face attendance system as shown in Fig. 1.

3.1. Face Capture from Webcam

Using the webcam the face capture happens and will be kept in the database. The photo that is taken will have a resolution of 358 x 358. These photos are used to store in the dataset.

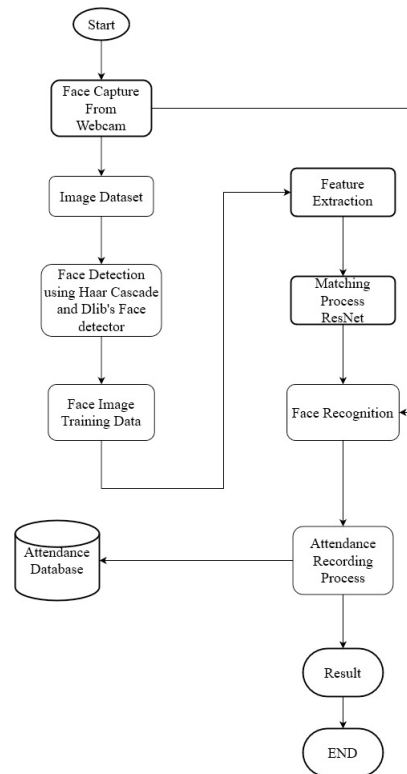


Fig. 1. The Proposed Model's Workflow

3.2. Image Dataset

For each person separate folders are created, and images are placed in that folder created for the person. Other images in the .jpg format also can be placed in the folder where the captured faces from the webcam are stored. These images are further used for preprocessing.

3.3. Face Detection

Once the dataset is created, the captured images are processed. The dlib frontal face detector uses the Haar cascade technique to locate the face region, which is then used to locate the 68 facial landmarks on the faces. The 68 landmarks that have been detected are shown in Fig. 2 [16] and Fig. 3 shows the landmarks located on the face.

3.4. Face Image Training data

The landmarks that are extracted from the face regions that the Haar cascade algorithm detects are saved as training data for the Resnet model and are subsequently utilised for feature extraction.

3.5. Feature Extraction

Using the ResNet algorithm first the features are extracted. The features extracted are the 128 unique face discriminators and these features are stored in a CSV file.



Fig. 2. Locations of the landmarks found.[17]

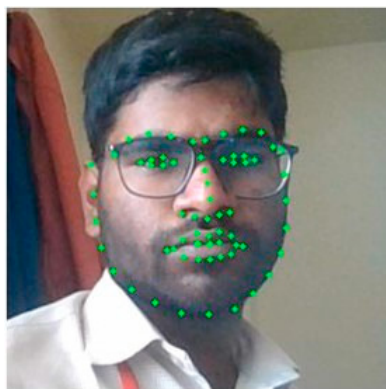


Fig. 3. A face's landmark locations

3.6. Matching Process

Using the 128 face discriminators and face region extracted the ResNet model is trained. The ResNet model has 50 layers. They include one fully connected layer, one initial convolutional layer, sixteen residual blocks (each consisting of several convolutional layers), and one global average pooling layer. [18][19]

3.7. Attendance Recording Process

The Haar cascade technique is once more used to recognise the face region of the camera-captured image. After that, the attendance is recorded and the face is matched using the 128 face discriminators.

3.8. Attendance Database

The attendance recorded in every class is stored in the SQLite database and to view the attendance of the date a user-friendly webpage has been created.

4. Evaluation Measures

The performance of the model is assessed using a confusion matrix. True positives (TP) indicate the number of faces correctly identified by the trained model. True negatives (TN) represent the number of faces classified as unknown that the model was not trained on. False positives (FP) refer to faces that the model incorrectly predicts as present in the dataset, while false negatives (FN) indicate the number of faces that the model fails to recognize, despite

their availability in the dataset. The accuracy, precision, and recall are calculated from the confusion matrix, with the corresponding formulas provided in Equations 1, 2, and 3.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

$$Precision = \frac{TP}{TP + FP} \quad (2)$$

$$Recall = \frac{TP}{TP + FN} \quad (3)$$

5. Implementation and Results

For the implementation of this attendance system, Python 3 has been used for the programming and for capturing of the faces the PC camera is used. First, the dataset has to be prepared. To prepare the dataset faces are captured using the web camera. In Fig. 4 we can see a red rectangle as the face is not in range to find the landmarks so we cannot capture the image that is out of range. The faces that are within the range can be captured and saved as shown in Fig. 5. Additionally, it enables users to clear old face data and add new faces.

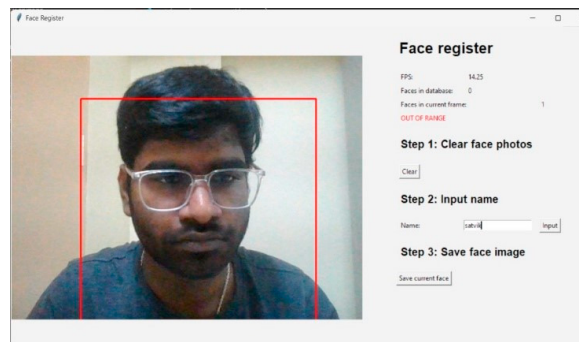


Fig. 4. Face capture from the web camera.

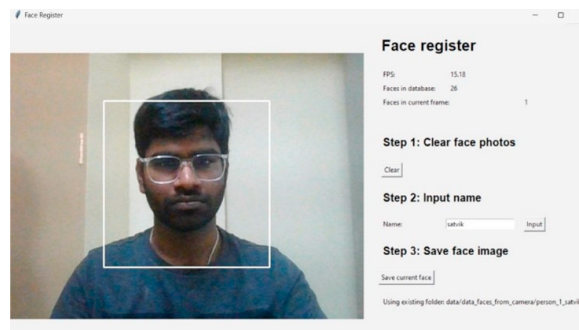


Fig. 5. Saving the Captured face to a predetermined location.

After the creation of the data the 68 landmarks of the face are found, and these are given as the input for the ResNet algorithm for extracting the features.

As shown in Table. 1 the 128 unique face-discriminating attributes are obtained and saved in the CSV file. The feature extraction is done by the convolutional layers that are present in the ResNet model. The features are generated by convolving the input images with learnable filters. These features are helpful in training the ResNet model.

Table 1. Features Extracted

Name	Feature-1	Feature-2		Feature-127	Feature-128
EAC21044	0.26019	0.1368	...	0.06647	-0.057
EAC21071	0.19541	0.1293	...	0.09931	0.0403
EAC21046	0.15161	0.1302	...	0.08106	-0.045
EAC21031	0.14779	0.1961	...	-0.0423	0.0104
...	...	...	...	...	...
EAC21030	0.22771	0.1371	...	-0.0200	-0.079
ECE21189	0.25036	0.1495	...	0.1012	-0.019
EAC21033	0.24270	0.1729	...	0.01048	-0.009

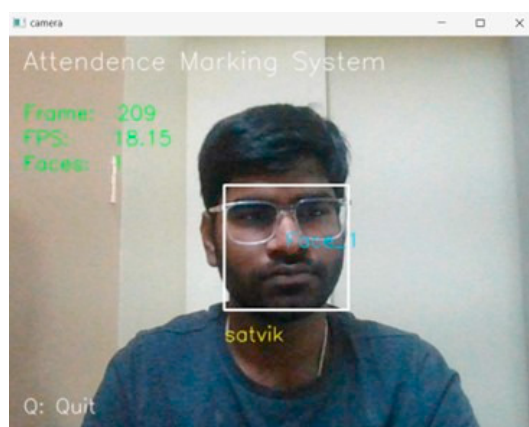


Fig. 6. Face recognition.

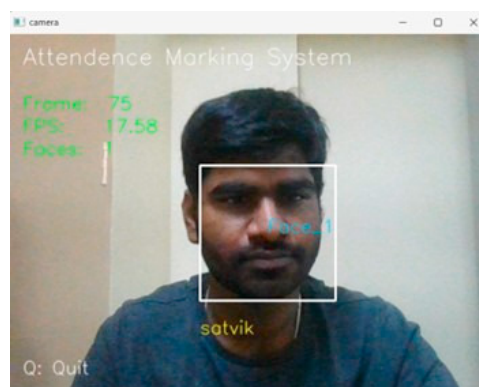


Fig. 7. Face recognition with occlusions.

After the feature extraction, the model is trained and saved. Now the faces are given as the input for the system to mark the attendance. As shown in Fig. 6 the face is identified and the person is marked present for that particular class. This model is also able to recognize the face under occlusion (a person without specs) as shown in Fig. 7.

The model will also recognize the multiple faces as shown in Fig. 8. It will mark the attendance of multiple people at the same time. Rather than capturing the faces from the camera, this model will also enable the user to give video input for marking the attendance.

A Flask web application is developed to display attendance data stored in an SQLite database. It defines two routes: 'attendance' to handle the form submission and retrieve attendance data based on the selected date, and '' to render the main page with a date selection form. To retrieve attendance records for the chosen day, the script establishes

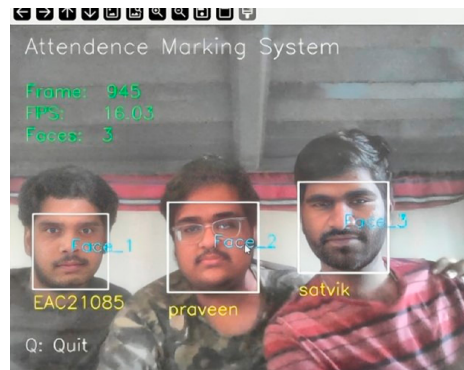


Fig. 8. Face recognition with multiple people.

a connection with the database and runs an SQL query. As shown in Fig. 9 & 10 the administrators can view the attendance report of any date.

Attendance Tracker Sheet	
Select Date: 02-Apr-2024 Show attendance	
Attendance Data Table	
Name	Time
satvik	19:34:02
EAC21085	19:34:48
proveen	19:36:03
EAC21031	20:52:23
EAC21003	20:52:45

Fig. 9. Attendance Report on 2nd April 2024.

Attendance Tracker Sheet	
Select Date: 06-Apr-2024 Show attendance	
Attendance Data Table	
Name	Time
satvik	08:24:00
EAC21033	08:24:38
27091	08:25:19
proveen	08:25:15
EAC21036	08:23:06

Fig. 10. Attendance Report on 6th April 2024.

The model is tested with 60 persons (50 people in the dataset and 10 people not in the dataset). The performance is tested under 2 conditions normal conditions Low lighting, and occlusion conditions (wearing glasses, headgear, or facial hair). Table. 2 a shows the results obtained in the normal conditions. In normal conditions accuracy, precision and recall are obtained as 100% as demonstrated in the fig 11.

Table. 2 b shows the confusion matrix obtained in low lighting and occlusion conditions. False positives are assigned a value of 2, false negatives a value of 3, and True positives a value of 45. As shown in the Fig 11 the results show that 91.6% accuracy, 93.8% precision, and 95.8% recall were attained.

Finally, the model is compared with the SOTA models. Many previous works that are discussed in the literature survey section did not mention the numerical values for the performance metrics. The works given the numerical values are compared in Table. 3. The results of the implemented model are good compared to the SOTA models. The SOTA models didn't focus on the occlusion conditions while the proposed model is performing better in the occlusion conditions as well.

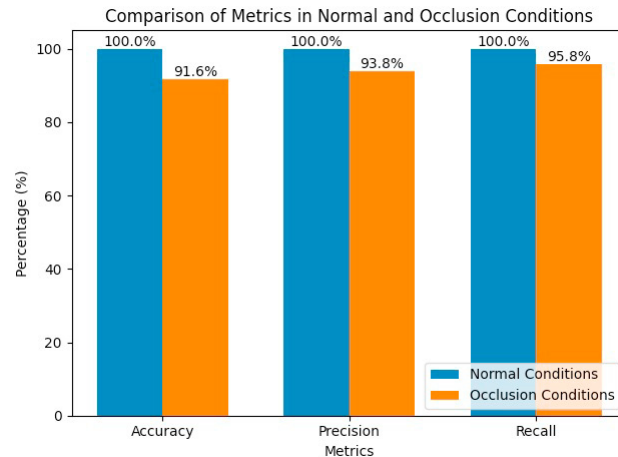


Fig. 11. Attendance Report on 2nd April 2024.

Table 2. Confusion Matrices: (a) Normal Conditions vs (b) Occlusion Conditions

	(a) Normal Conditions		(b) Occlusion Conditions	
	Predicted Not Present	Predicted Present	Predicted Not Present	Predicted Present
Actual Not Present	10 (TN)	0 (FP)	9 (TN)	3 (FP)
Actual Present	0 (FN)	50 (TP)	2 (FN)	46 (TP)

Table 3. Comparasion of Results with the State of the Art

Author(Year)	Model	Accuracy
S.Matilda et al.(2019)[5]	Haar Cascade	99.38%
S.Bhattacharya et al.(2018)[20]	CNN	99.4%
Lin zhi-heng et al.(2019)[21]	Neural Networks	85%
Syam Kakarla et al.(2020)[13]	CNN	99.4%
Azhaguraj R et al.(2022)[22]	Haar Cascade, LBPH	85%
Vidya Patil et al.(2020)[23]	SVM	95%
Senthil G.A et al.(2024)[2]	KNN	94.32%
J srikanth et al. (2024)[3]	Deep Face Learning	95%
Ravishka F et al. (2024)[4]	LPBH	99.22%
Proposed Model (2024)	Resnet, Haar Cascade	100% (Normal Conditions) 91.6% (Occlusion Conditions)

6. Conclusion and Future Scope

The suggested method uses cutting-edge facial recognition technology to effectively automate the attendance marking procedure. The system achieves excellent accuracy in a variety of difficult situations, such as dim lighting, occlusions, and a wide range of facial emotions, by combining the deep learning architecture of ResNet with the reliable facial identification capabilities of Haar Cascade. As a result, the model can be adjusted to real-world settings with less-than-ideal conditions. By facilitating secure storage, fast data access, and simple attendance record retrieval—even for big datasets—the utilization of a SQLite database improves system efficiency. Furthermore, the system’s scalability for larger institutions or corporate settings is demonstrated by its capacity to handle several faces at once without sacrificing accuracy. The experimental findings validate the model’s appropriateness for real-world implementation in corporate, educational, and secure access control settings by confirming that it functions well in both standard and occlusion scenarios.

Future improvements can concentrate on strengthening the model's resilience by adding object-tracking techniques or Kalman Filtering to manage dynamic settings and track numerous faces at once. In order to provide deeper insights into attendance trends, another extension entails integrating cloud-based solutions for scalable data administration and real-time analytics. Furthermore, creating a mobile application for attendance tracking and remote access could improve user experience and system accessibility. By using multi-factor authentication based on biometrics, security may be improved and attendance data can be kept safe. Improvements to facial recognition algorithms can increase accuracy even more in difficult situations, such as partial occlusions. Finally, by incorporating machine learning into adaptive model training, the system would be able to adapt to changing user needs and new data over time.

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